Healthcare Data Analytics Project

A comprehensive analysis of hospital performance across multiple cities

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Tools Used:

Python | SQL | Power BI

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Contents

1. Introduction

- This project focuses on analyzing healthcare data to understand patient trends, revenue generation, and treatment costs across multiple cities.
- The main objective of this project is to extract meaningful insights using data analysis tools like Python, SQL, and Power BI.

2. Dataset Description

- The dataset used in this project is a healthcare dataset. The dataset contains approximately 500 unique patient records, verified using the nunique() function in Pandas.
- It includes the following columns:
  - Patient ID
  - Age
  - City
  - Disease
  - Admission Date
  - Treatment Cost
  - Payment Status
  - Insurance Cover

3. Data Cleaning and Processing

- Data cleaning was performed using Python libraries such as Pandas and NumPy.
- Missing values were handled, and new features were created, such as:
  - Age Group
  - Final Bill Amount
- This step ensured that the dataset was ready for analysis.

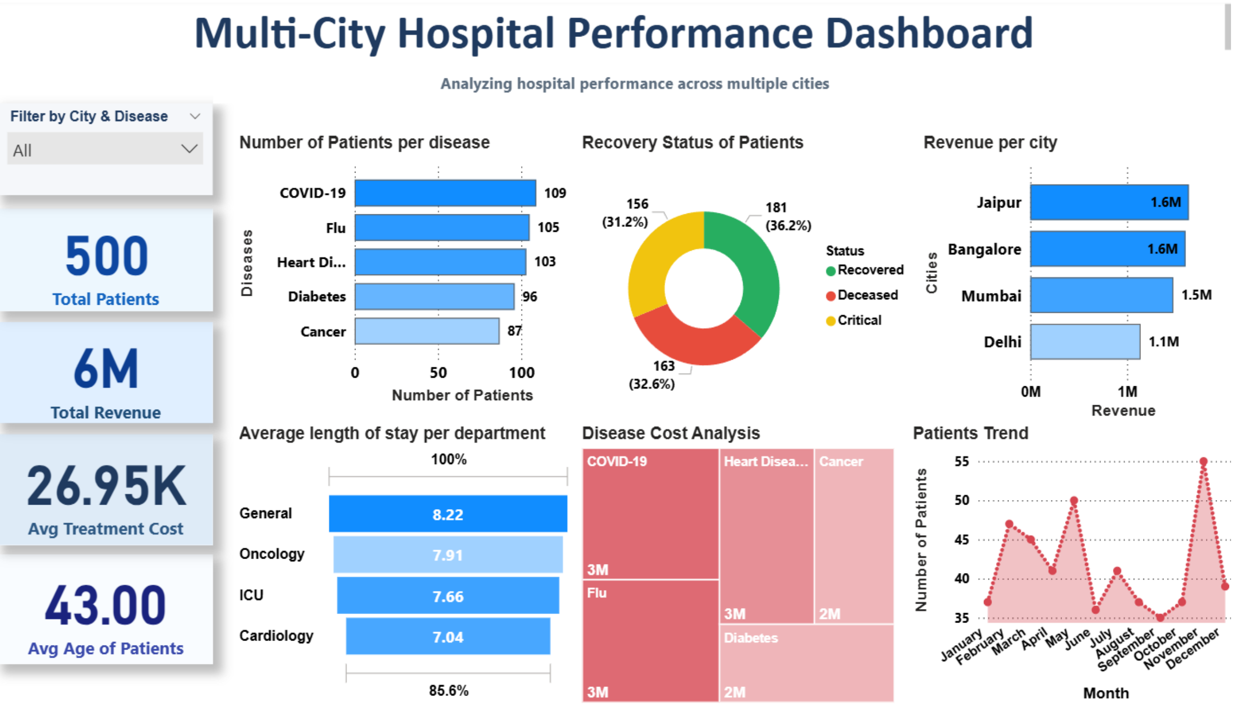
4. Data Analysis

- Data analysis was performed using Python and SQL. Python was used for initial exploration and transformation, while SQL was used to filter and extract meaningful insights from the processed dataset.

5. SQL Integration

- The processed dataset was connected to MySQL using Python libraries such as SQLAlchemy and PyMySQL.
- SQL queries were used to analyze the data further, including filtering by city, disease, and payment status.

6. Data Visualization



An interactive dashboard was created using Power BI to visualize key insights such as revenue by city, disease distribution, and patient trends.

7. Key Insights

- Certain cities contribute higher revenue compared to others
- Some diseases are associated with higher treatment costs
- Patient inflow varies across different months
- A portion of payments remains unpaid, indicating potential revenue gaps

8. Conclusion

- This project helped in understanding how data analytics can be used in the healthcare sector to derive meaningful insights.
- This project enhanced my understanding of applying appropriate analytical techniques and tools to solve real-world problems effectively.

9. Future Improvements

- Use real-world healthcare datasets for more accurate analysis
- Implement predictive models to forecast patient trends
- Deploy the dashboard for real-time monitoring